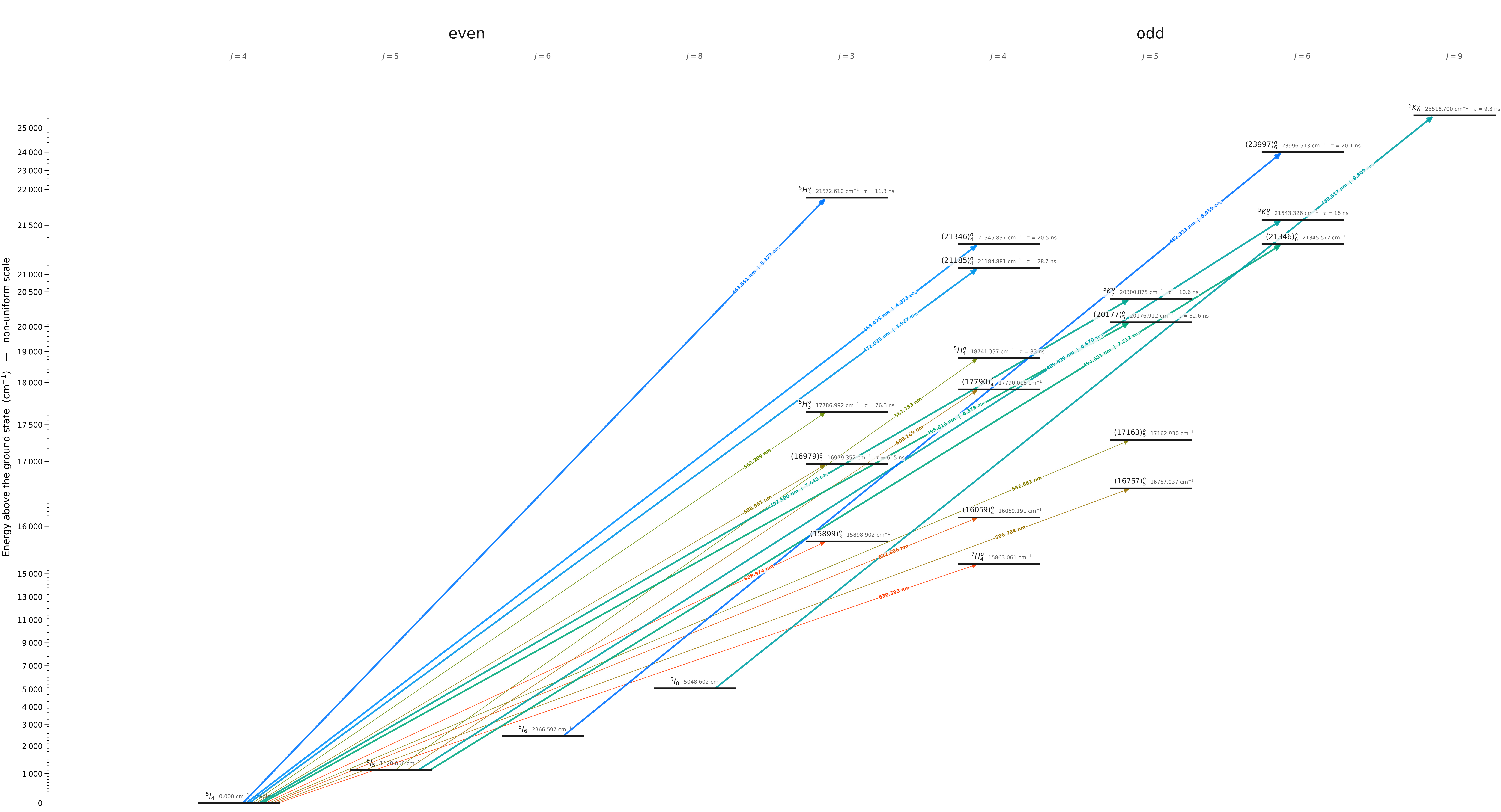
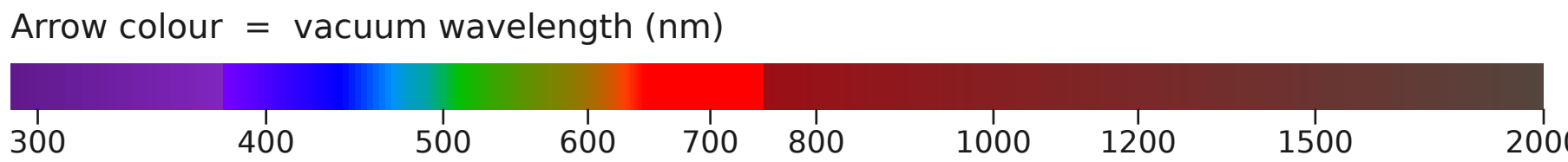


Neodymium

Nd (Nd I). Strongest classified transitions below 2 μm (NIST): 22 levels, 18 lines.

Nd⁺ ionisation limit 44560.11 cm⁻¹ = 5.52475 eV (224.416 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $| \langle J || e r || J' \rangle |$ in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, *n* = 70.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	⟨J er J'⟩ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
⁵ I ₄ - ⁵ H ₃	562.2086	562.0526	533.2406	-	-	2.086	-	first step of a three-step photoionization scheme propose...

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

¹⁴³Nd hyperfine structure (*I* = 7/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
⁵ I ₄	-195.649(9)	122.25(28)	F=1/2: +3483.89 F=3/2: +3174.98 F=5/2: +2663.25 F=7/2: +1953.37 F=9/2: +1051.90 F=11/2: -32.75 F=13/2: -1290.27 F=15/2: -2708.52

Measured A, B constants; sources in the data file.

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
⁵ I ₄ - ⁵ H ₃	562.209	142-150	-3048.00
⁵ I ₄ - ⁵ H ₃	562.209	144-150	-2448.00
⁵ I ₄ - ⁵ H ₃	562.209	146-150	-1833.00
⁵ I ₄ - ⁵ H ₃	562.209	148-150	-1131.00
⁵ I ₄ - ⁵ H ₃	562.209	143-150	-3060.00
⁵ I ₄ - ⁵ H ₃	562.209	145-150	-2300.00
⁵ I ₅ - ⁵ H ₄	567.753	144-142	657.5(6)
⁵ I ₅ - ⁵ H ₄	567.753	146-144	586.9(5)
⁵ I ₅ - ⁵ H ₄	567.753	148-146	720.7(6)
⁵ I ₅ - ⁵ H ₄	567.753	150-148	1169.4(8)
⁵ I ₅ - ⁵ H ₄	567.753	143-142	257.6(5)
⁵ I ₅ - ⁵ H ₄	567.753	145-144	190.2(4)
⁵ I ₄ - (17163) ₅	582.651	144-142	-262.2(9)
⁵ I ₄ - (17163) ₅	582.651	146-144	-265.8(6)
⁵ I ₄ - (17163) ₅	582.651	148-146	-228.5(5)
⁵ I ₄ - (17163) ₅	582.651	150-148	-127.10(20)
⁵ I ₄ - (17163) ₅	582.651	143-142	-146.7(7)
⁵ I ₄ - (17163) ₅	582.651	145-144	-155.2(6)
⁵ I ₄ - (16979) ₃	588.951	144-142	-223.7(3)
⁵ I ₄ - (16979) ₃	588.951	146-144	-227.1(3)
⁵ I ₄ - (16979) ₃	588.951	148-146	-194.4(3)
⁵ I ₄ - (16979) ₃	588.951	150-148	-105.80(20)

Shift = ν(first isotope) - ν(second isotope).